How one reversible block update works

Exact entries from the frozen 256-state permutation

A Four cell states

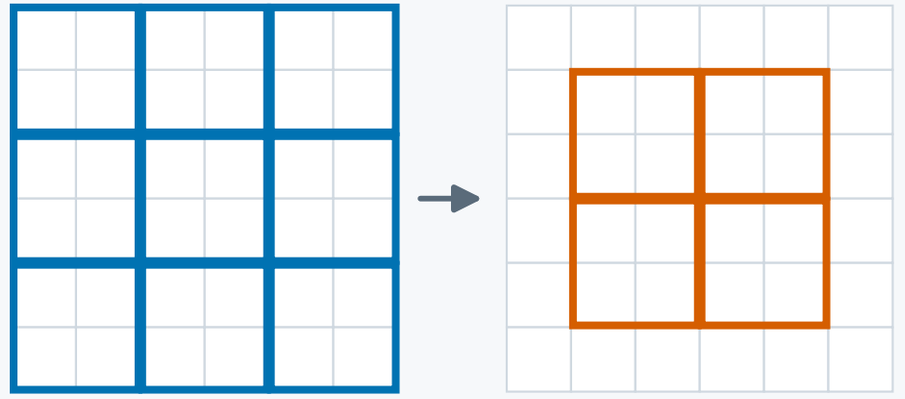
0 — **empty**
Q = 0, M = 0

+ — **positive**
Q = +1, M = 1

− — **negative**
Q = −1, M = 1

b — **bound-neutral**
Q = 0, M = 1

B Alternating partitions



C Transport-like lookup

code 1

+	0
0	0

→

code 4

0	+
0	0

Q: +1 → +1 **M: 1 → 1**

transport: Q and M unchanged

D Balanced reaction lookup

code 130

−	0
0	−

→

code 154

−	−
+	−

Q: -2 → -2 **M: 2 → 4**

balanced pair creation: Q unchanged, M + 2